PAOLO MIGUEL C. MORATO

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SUMMARY

Full-stack developer with 2+ years shipping production web apps and leading AI-driven automation that cut costs by 95%. Strong across React/TypeScript, Node.js/Python, CI/CD (Docker, Jenkins), and applied LLM tooling — OpenAI API, retrieval-augmented pipelines.

WORK EXPERIENCE

DocOps Engineer — *ITRS Phils Inc.* Sep 2024 – Present

- Build and maintain a Hugo-based documentation website and React/TypeScript web applications — shipping new features, resolving bugs, and improving accessibility.
- Led the migration of documentation translation to an AI-driven system, automating English→French/German/Dutch translation and reducing costs by 95% while maintaining near human-level quality.
  - Designed a retrieval-augmented, semantic few-shot translation pipeline (embedding-based retrieval of human-translated examples feeding in-context GPT translation) and evaluated quality using BLEU, chrF, BERTScore, and COMET.
  - Built the end-to-end system with PySide6 for the desktop GUI and OpenAI SDK, Sentence Transformers, PyTorch, NumPy, and scikit-learn for the ML backend.
- Built and maintain CI/CD pipelines with Docker and Jenkins to support fast, reliable deployments.
- Raised search indexing accuracy from 20% to 80% by automating sitemap URL generation and removing outdated pages from the index.
- Manage Google Analytics 4 and Google Search Console to monitor and optimize site performance and SEO.
- Collaborate cross-functionally with web developers, DevOps engineers, and technical writers to ship documentation website solutions.

Full Stack Web Development Intern — *iBuild.PH* Aug 2023 – Nov 2023

- Developed and maintained both the frontend and backend of an eCommerce website for a music ministry.
- Implemented product display and online payment features, improving user experience and site functionality.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, Java, HTML, CSS/LESS

Frontend: React.js, Vite, Hugo

Backend: Node.js, Express.js, FastAPI, Flask

DevOps & Cloud: Docker, Jenkins, CI/CD, GCP (Compute Engine, Cloud Run, Cloud Storage), AWS (EC2, Lambda)

AI / ML & Automation: OpenAI API, RAG pipelines, Sentence Transformers, PyTorch, scikit-learn, Cursor (AI-assisted development)

Databases: PostgreSQL, MySQL, MongoDB

Other: Jira, Google Analytics 4, Google Search Console, Drupal

PERSONAL PROJECTS

FloodWatcher — Forecasting Marikina River Level with GPT + Kernel PCA

- Co-authored a published paper introducing a GPT model with parameter embeddings pre-trained via Kernel PCA to forecast Marikina River water levels, capturing linear and non-linear relationships between meteorological and hydrological data; outperformed a traditional Transformer baseline. Secured 2nd place in a research competition.

Stack: *Python, GPT / Transformer architectures, Kernel PCA, React, Flask*

Paper: *https://www.sciencedirect.com/science/article/pii/S1877050924030540*

Pulsecraft — Local AI Voice Generation Studio

- Built a local AI voice generation studio for reusable voice profiles, reference-audio transcription, and cloned voice clip generation via a web dashboard connected to a local FastAPI engine.

Stack: *React, TypeScript, Vite, FastAPI, Python, SQLAlchemy, SQLite, Qwen TTS, Whisper*

Barangay Management and Request System (BMRS)

- Built a PERN stack web application enabling residents to request barangay certificates and IDs online, improving accessibility and processing efficiency.

Stack: *PostgreSQL, Express.js, React, Node.js*

EDUCATION

Bachelor of Science in Computer Science 2020 – 2024

Awards: Cum Laude

Science, Technology, Engineering and Mathematics (STEM) 2018 – 2020

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